

NO ALTERNATING PLANE GRAPH HAS EXACTLY TWO VERTEX DEGREES, OR EXACTLY TWO FACE SIZES

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ABSTRACT. In an alternating plane graph, adjacent vertices have different degrees and adjacent faces have different sizes. We prove the conjecture of Althöfer, Haugland, Scherer, Schneider and Van Cleemput that no such graph has exactly two degrees, and none has exactly two face sizes. Give every edge the reciprocals of its two end degrees and its two face sizes. These charges sum to $e + 2$. Two values on one side make the graph bipartite on that side, which forces even values on the other side, and then no edge gets more than 1. The result still holds when bridges are allowed; there a large face on each bridge pays for the few edges that can exceed 1.

1. INTRODUCTION

An alternating plane graph is a plane graph in which neighbours differ: adjacent vertices have different degrees, and adjacent faces have different sizes. How few degrees and sizes can such a graph get away with? Althöfer, Haugland, Scherer, Schneider and Van Cleemput [1], who introduced these graphs, conjectured that the answer is three of each:

Conjecture 10.1. There are no $2, Y$ -alternating plane graphs and no $X, 2$ -alternating plane graphs.

Here an X, Y -alternating plane graph has exactly X distinct degrees and exactly Y distinct face sizes. Two is the natural first case: alternation on vertices alone already forces at least two degrees.

They showed that a $2, Y$ -alternating plane graph has degrees 3 and 4, or 3 and 5, with at least 25 or 56 vertices respectively, beyond the reach of their `plantri` [2] search. For weak alternating plane graphs, which allow degree 2, they bounded the number of faces by summing $1/r + 1/s$ over edges between an r -face and an s -face.

That face count never uses degree 2, and it is most of the proof. Add the same count over the two ends of each edge. By Euler's formula the four reciprocals, summed over all edges, give $e + 2$, so some edge carries more than 1. Alternation alone allows $\frac{1}{3} + \frac{1}{4}$ on each side, which is $\frac{7}{6}$ and too much. But two values on one side make the graph bipartite there, so the values on the other side are even. Distinct even values are at least 4 and 6, and $\frac{1}{4} + \frac{1}{6} = \frac{5}{12}$. Now $\frac{7}{12} + \frac{5}{12} = 1$ at every edge, a contradiction.

We prove the conjecture in §4 (Theorem 4.1). In §5 we allow bridges, requiring face alternation only between distinct faces, and show that both statements survive. This is the harder part.

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This charge belongs to the tradition of Lebesgue [6] and Kotzig [5], now the discharging method [3, 4]. Here almost nothing is discharged.

2. PRELIMINARIES

All graphs are finite, simple, plane and connected, and faces include the unbounded one.

Definition 2.1 ([1], Definition 2.1). An *alternating plane graph* is a connected plane graph in which (a) every vertex has degree at least 3; (b) every face has size at least 3; (c) adjacent vertices have different degrees; (d) adjacent faces have different sizes. An *X, Y -alternating plane graph* is one with exactly X distinct vertex degrees and exactly Y distinct face sizes.

Convention 2.2 (face size). The size of a face is the length of its boundary walk [7, §3.3]. A bridge lies on one face from both sides and counts twice towards its size.

In a bipartite plane graph every closed walk is even, so every face has even size under this convention. Counting distinct boundary vertices would lose that.

Convention 2.3 (bridges). In (d), the faces on the two sides of an edge must have different sizes, so a bridge breaks (d) and alternating plane graphs have none [1]. A plane graph is *bridge-relaxed alternating* if it satisfies (a)–(c), and (d) for distinct adjacent faces.

3. THE EDGE CHARGE

Lemma 3.1. *Let G have v vertices, $e \geq 1$ edges and f faces. For an edge $a = uw$ with faces $F^-(a)$ and $F^+(a)$ on its two sides, put*

$$c(a) = \frac{1}{\deg u} + \frac{1}{\deg w} + \frac{1}{|F^-(a)|} + \frac{1}{|F^+(a)|}.$$

Then $\sum_a c(a) = v + f = e + 2$. Equivalently the excesses $x(a) = c(a) - 1$ satisfy

$$(1) \quad \sum_a x(a) = 2.$$

Proof. Each vertex hands $1/\deg$ to each edge end at it, and each face hands $1/|F|$ to each edge side on its boundary walk. Each pays out exactly 1, so the edges receive $v + f$, which is $e + 2$ by Euler's formula. $\square$

A bridge has $F^-(a) = F^+(a)$ and receives $2/|F|$ from that face. So the lemma holds with bridges too, and §5 uses it that way.

Example 3.2. Replace each side of a triangle by two paths of length two. This gives 9 vertices, 12 edges and 5 faces. Every edge joins degree 2 to degree 4 and separates a quadrilateral from a hexagon, so $c(a) = \frac{7}{6}$ and the 12 excesses sum to 2. Only the degree-2 vertices keep it from being alternating. It also attains the face bound $f \leq \frac{5}{12}e$ of [1, (9.5)].

The second ingredient is parity. At a vertex u of degree d , list the edges in cyclic order $a_1, \dots, a_d$, and call the region between a_i and a_{i+1} a *corner*. The two corners beside a_i lie in the same face exactly when a_i is a bridge.

Lemma 3.3 (parity). *Suppose the faces of G have a 2-colouring that is proper on distinct adjacent faces. Then $\deg u$ minus the number of bridges at u is even, for every vertex u .*

Proof. Going once round u , the colour changes across each non-bridge edge and stays the same across each bridge. We return to the first colour, so the number of changes is even. $\square$

With no bridges, every degree is even. In an $X, 2$ -alternating graph the two face sizes give the colouring.

4. CONJECTURE 10.1

Theorem 4.1. *There is no $2, Y$ -alternating plane graph and no $X, 2$ -alternating plane graph.*

The two cases exchange vertices and faces (Table 1). They are not formally dual. Duality needs 3-edge-connectivity [1, Lemma 2.2], since a 2-edge cut gives parallel edges in the dual. So we prove both.

	vertex side	face side	$c(a)$
two degrees	$\leq \frac{1}{3} + \frac{1}{4}$	$\leq \frac{1}{4} + \frac{1}{6}$	≤ 1
two face sizes	$\leq \frac{1}{4} + \frac{1}{6}$	$\leq \frac{1}{3} + \frac{1}{4}$	≤ 1

TABLE 1. Bounds on the two halves of $c(a)$ in the two cases of Theorem 4.1.

Proof. *Two degrees* $3 \leq d_1 < d_2$. By (c) every edge joins a d_1 -vertex to a d_2 -vertex. So the vertex side of $c(a)$ is at most $\frac{1}{3} + \frac{1}{4}$, and G is bipartite. Every face then has even size, at least 4. The two faces at an edge are distinct, since there are no bridges, and by (d) their sizes differ. So they are at least 4 and 6, and the face side is at most $\frac{1}{4} + \frac{1}{6}$. Hence $c(a) \leq 1$ for every edge, against (1).

Two face sizes $s_1 < s_2$. By (d) every edge separates an s_1 -face from an s_2 -face, so the face side is at most $\frac{1}{3} + \frac{1}{4}$. The sizes colour the faces properly and there are no bridges, so every degree is even by Lemma 3.3, hence at least 4. Adjacent degrees differ, so the vertex side is at most $\frac{1}{4} + \frac{1}{6}$. Again $c(a) \leq 1$ everywhere. $\square$

The first case is Althöfer et al.'s bound [1, (9.9)] with the $\frac{1}{2}$ from degree 2 replaced by $\frac{1}{3}$.

5. ALLOWING BRIDGES

A bridge breaks both halves of the argument. Its two sides are one face, and its ends need not have even degree. What rescues it is that a bridge sits on a large face.

Lemma 5.1. *If G has minimum degree at least 3, every face containing a bridge has size at least 8.*

Proof. Let $a = uw$ be a bridge, and let $A \ni u$ and $C \ni w$ be the components of $G - a$. In A , u has degree at least 2. Let W_A be the boundary walk of A through the corner left by deleting a . It cannot have length 1 (a loop) or 2 (a double edge, or $A = K_2$, but $\deg_A u \geq 2$). So $|W_A| \geq 3$, and likewise $|W_C| \geq 3$. The face of G at a runs along W_A , across a , along W_C and back across a , so its size is $|W_A| + |W_C| + 2 \geq 8$. $\square$

Proposition 5.2. *There is no bridge-relaxed alternating plane graph with exactly two degrees.*

Proof. As before, G is bipartite with degrees $3 \leq d_1 < d_2$, and every face has even size at least 4. A non-bridge edge still has $c(a) \leq 1$. A bridge has $c(a) \leq \frac{1}{3} + \frac{1}{4} + \frac{2}{8} = \frac{5}{6}$ by Lemma 5.1. So (1) fails. $\square$

Theorem 5.3. *There is no bridge-relaxed alternating plane graph with exactly two face sizes.*

Proof. Let G have face sizes $s_1 < s_2$ and B bridges. If $B = 0$, Theorem 4.1 applies, so let $B \geq 1$.

Step 1: the bad edges. A face with a bridge has size at least 8, so $s_2 \geq 8$. Table 2 bounds $c(a)$, using that end degrees are distinct and at least 3. Only a non-bridge edge with an end of degree 3 can have positive excess. Call such an edge *bad*, and let P count them.

edge a	terms	$c(a) \leq$	$x(a) \leq$
bridge	$\frac{1}{3} + \frac{1}{4} + \frac{2}{8}$	$\frac{5}{6}$	$-\frac{1}{6}$
non-bridge, both ends of degree ≥ 4	$\frac{1}{4} + \frac{1}{5} + \frac{1}{3} + \frac{1}{8}$	$\frac{109}{120}$	$-\frac{11}{120}$
non-bridge, an end of degree 3	$\frac{1}{3} + \frac{1}{4} + \frac{1}{3} + \frac{1}{8}$	$\frac{25}{24}$	$\frac{1}{24}$

TABLE 2. Bounds on the charge of an edge in Theorem 5.3, where $s_1 \geq 3$ and $s_2 \geq 8$.

Step 2: each bridge takes at most two bad edges. This is the hard step. Let a be bad, with u its end of degree 3. By Lemma 3.3 the number of bridges at u is odd. It is not 3, because a is not a bridge. So u lies on exactly one bridge, and we assign a to it (Figure 1).

Now fix a bridge b . Every bad edge assigned to b has its degree-3 end on b . By (c) at most one end of b has degree 3, and at that end only two edges other than b remain. So $P \leq 2B$.

Step 3: the sum. Every other edge has $x(a) \leq 0$. So by (1) and Table 2,

$$(2) \quad 2 = \sum_a x(a) \leq \frac{P}{24} - \frac{B}{6} \leq \frac{2B}{24} - \frac{B}{6} = -\frac{B}{12} < 0.$$

$\square$

Step 2 has room to spare: $P \leq 4B$ would already give a sum of at most 0.

Remark 5.4. Bridge-relaxed alternating plane graphs with bridges do exist. Take three copies of the 17-vertex (3, 4, 5)-alternating plane graph of [1, Figure 2], each drawn with a degree-5 vertex z_i on the outer face, and join z_1 and z_3 to z_2 by

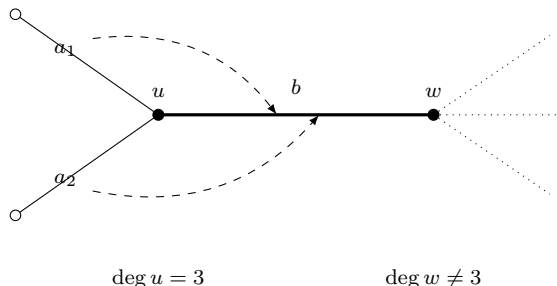


FIGURE 1. The worst case at a bridge $b = uw$: both other edges a_1, a_2 at the degree-3 end u are bad, and both are assigned to b (dashed arrows).

bridges. Then $\deg z_2 = 7$, $\deg z_1 = \deg z_3 = 6$, and the merged outer face has size at least 13, larger than every inner face. It has more than two degrees and more than two face sizes.

6. WHAT STAYS OPEN

The charge sums to a constant, so it says nothing about proportions. Althöfer et al. also ask how the degrees of large $(3, 4, 5)$ -alternating plane graphs are distributed: the ratio of degree-4 to degree-3 vertices lies in $[1, 1.5]$ asymptotically, and they ask whether it has a limiting density there. A per-edge bound cannot see that. We do not know what can.

DISCLOSURE OF AI USE

Large language models were used throughout this work. They helped search for arguments, reviewed the proofs adversarially, and drafted the prose from the author's notes; the author edited it. The idea of allowing bridges (§5) came from an AI reviewer. The systems were Claude Opus 5 and Claude Fable 5.1 (Anthropic), ChatGPT at its Pro tier and Codex **gpt-5.6-terra** and **gpt-5.6-sol** (OpenAI), and DeepSeek **v4-pro** and **v4-flash**, used from February to September 2026. The author checked the mathematics and is responsible for it. No AI system is an author.

COMPUTATIONS

The proofs use no computation. Exact-arithmetic checks of the inequalities and machine checks of explicit examples are archived at [doi:10.5281/zenodo.22269200](https://doi.org/10.5281/zenodo.22269200).

REFERENCES

- [1] I. Althöfer, J. K. Haugland, K. Scherer, F. Schneider and N. Van Cleemput, *Alternating plane graphs*, *Ars Math. Contemp.* **8** (2015), 337–363. doi:10.26493/1855-3974.584.09a.
- [2] G. Brinkmann and B. D. McKay, *Fast generation of planar graphs*, *MATCH Commun. Math. Comput. Chem.* **58** (2007), 323–357.
- [3] D. W. Cranston and D. B. West, *An introduction to the discharging method via graph coloring*, *Discrete Math.* **340** (2017), 766–793. doi:10.1016/j.disc.2016.11.022.
- [4] S. Jendrol' and H.-J. Voss, *Light subgraphs of graphs embedded in the plane — a survey*, *Discrete Math.* **313** (2013), 406–421. doi:10.1016/j.disc.2012.11.007.

- [5] A. Kotzig, *Contribution to the theory of Eulerian polyhedra*, Mat.-Fyz. Časopis Slovensk. Akad. Vied **5** (1955), 101–113.
- [6] H. Lebesgue, *Quelques conséquences simples de la formule d'Euler*, J. Math. Pures Appl. (9) **19** (1940), 27–43.
- [7] B. Mohar and C. Thomassen, *Graphs on Surfaces*, Johns Hopkins University Press, Baltimore, 2001.

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